

DJAMEL BOUGOUFFA

Full-Stack Software Engineer | AI & Agentic Tooling | Rust, TypeScript

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SUMMARY

Full-stack software engineer with 7+ years taking products from idea to production, for client teams and on my own. Core stack TypeScript, React, Node.js and Java, with recent work centred on Rust developer tooling, MCP servers and RAG pipelines adopted by 110+ engineers. Remote-first and autonomous: I own features end to end, publish open source on crates.io, and write about the craft in English and French.

SKILLS

Languages: TypeScript, JavaScript, Rust, Java, SQL, PHP, Bash, HTML5, CSS3

Frontend: React, Next.js, Angular, React Native (Expo), Redux, Atomic Design, Micro Frontends

Backend: Node.js, Express, NestJS, Spring Boot, GraphQL, REST, WebSocket, RabbitMQ

AI & Tooling: MCP servers and clients, RAG, agentic workflows, Tree-sitter/AST parsing, embeddings, vector search (HNSW), Claude Code

Databases: PostgreSQL, MySQL, MariaDB, MongoDB, Redis

Cloud & DevOps: Docker, Docker Compose, Kubernetes, Swarm, Traefik, GitHub Actions, Jenkins, CI/CD, GCP

Practices: Agile/Scrum, TDD, code review, hexagonal architecture, modular monorepo, async remote collaboration

Spoken Languages: French (native), English (full professional proficiency)

EXPERIENCE

Ippon Technologies, Paris, France (*Full-Stack Software Engineer*)

Nov 2025 - Present

Consulting and innovation firm, split between internal R&D on AI developer tooling and delivery on client platforms.

- Co-created a proprietary AI-assisted legacy modernization platform (owned by the company), taking it from R&D prototype to a validated product adopted by **110+ consultants**.
- Built the platform across three surfaces sharing one core: a **Rust** CLI, a web frontend and an **MCP server**.
- Designed the **RAG** and **Tree-sitter** pipelines behind it: AST-level chunking, embeddings and retrieval that generate code documentation automatically.
- Implemented agentic workflows over MCP, connecting IDE assistants to testing, refactoring and code exploration.

Client engagement: Suez IWS

Mar 2026 - Present

- Rebuilt the **CI/CD pipeline** of a legacy B2B extranet handling regulated hazardous-waste workflows, and contributed to its security hardening.
- Migrated a legacy **Liferay** frontend toward **Angular micro frontends** to improve modularity and maintainability.
- Developed backend features in **Java** within the legacy application core, alongside the frontend modernization effort.

Go-Together, independent product (*Full-Stack Engineer, solo*)

May 2025 - Oct 2025

Intercity carpooling product for Thailand, built solo end to end. Full-time until October 2025, continued as a side project since.

- Shipped the product end to end: a **NestJS** API (authentication, ride matching, bookings), **React Native** apps for riders and drivers, a **Next.js** web app and an **Astro** marketing site in English and Thai.
- Designed a transparent pricing engine and integrated **PromptPay** QR payments, the dominant instant-payment system in Thailand.
- Ran market research and field prospection in Thailand, and took the product to a pre-launch waitlist at gotogother.com.

SYSTRAN by ChapsVision, Paris, France (*Full-Stack Software Engineer*)

Jan 2025 - Apr 2025

AI-powered NLP platform specialized in enterprise translation.

- Developed product features in **TypeScript**, **React** and **Next.js**, and maintained the **Node.js** and **Express** services behind them.
- Proposed modernization directions for a legacy codebase to improve maintainability and delivery quality.

Career break (*Independent travel across Asia*)

Mar 2024 - Jan 2025

- Took an intentional break to travel across Southeast Asia, which seeded the Go-Together product above.

Wemanim Group, METRO France, Paris, France (*Full-Stack Software Engineer*)

Mar 2022 - Mar 2024

Consulting role on large-scale retail and operational platforms.

- Led the frontend repository of a bonus-by-objectives product serving **4,000+ employees and 1,000 managers**.
- Delivered point-of-sale, self-service kiosk and order-tracking features running across **~95 warehouses** in France.
- Worked with React, Node.js, TypeScript, Redux and Redis, applying **hexagonal architecture** and **Atomic Design**.

Ma Formation Médicale, Paris, France (*Full-Stack Software Engineer*)

Sep 2020 - Feb 2022

Platform dedicated to medical education and training.

- Migrated a legacy **PHP** codebase toward modular **JavaScript** services and reshaped the API architecture.
- Rebuilt the backend around a component-based approach to improve maintainability and scalability.

Fokal Studio, Ferrières-en-Brie, France (*Full-Stack Software Engineer*)

Jun 2018 - May 2020

Digital agency specialized in custom websites and digital solutions.

- Delivered custom websites and CMS solutions for a portfolio of agency clients.

OPEN SOURCE & PROJECTS

rustkit-ai / semtree *On-device semantic code intelligence, open source in Rust* github.com/rustkit-ai/semtree

- Built a Rust library, CLI and **MCP server** that parse, embed and search any codebase by meaning, fully on-device, with no API key and no daemon.
- Combined **Tree-sitter** parsing, local **fastembed** embeddings (ONNX on CPU) and an **HNSW** index, with hybrid ranking that fuses vector similarity and BM25.
- Kept the design model-agnostic behind **Embedder** and **VectorStore** traits, so a local **Ollama** model or a remote API swaps in without touching the core.
- Published 8 crates to crates.io under MIT with CI and docs.rs, **1,700+ cumulative downloads**, current release v0.5.0.

djamel-bougouffa.com *Engineering blog and portfolio, written in English and French* djamel-bougouffa.com

- Wrote and shipped **11 articles** in both English and French, on Rust, MCP servers, AI-assisted workflows and software craft.
- Built and run the site end to end on **Astro** and Netlify: bilingual routing, structured data for search, and RSS syndication to dev.to and Medium.

EDUCATION

Master's Degree in Computer Engineering, Expert Web Architect, Institut F2I, Paris 2021

Bachelor's Degree in Computer Engineering, Digital Project Manager, Institut F2I, Paris 2020

Diploma of Advanced Technician, Solutions and Business Applications, IRIS, Paris 2018